

Fig. 2: Actual-size PCB layout of the electronic thermostat circuit

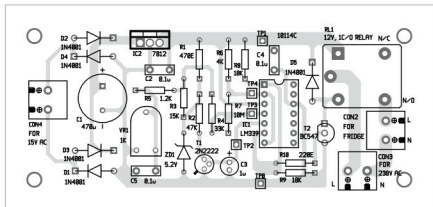


Fig. 3: Component layout of the PCB

inverting input is greater than that at non-inverting input, comparator output pin 2 is high, T2 is saturated and energises the relay coil and the appliance remains switched on.

Off state. When temperature goes below -15°C , comparator output goes low, relay gets de-energised and the appliance goes into off state. When comparator goes low, feedback voltage of 20mV vanishes and voltage at the inverting input corresponds to a temperature of -5°C . So the temperature cycles between -5°C and -15°C .

Reference setting using VR1 decides the less negative of the two temperature extremes and the hysteresis depends on the voltage divider arrangement of R7 and R4.

Let 'T' be the desired temperature hysteresis, then,

$$T (^{\circ}\text{C}) = (3000 \times R4) / (R4 + R7)$$

Power supply. DC supply for

the circuit including the relay consists of a conventional bridge rectifier, step-down transformer X1 and filter capacitor C1, along with 3-terminal voltage regulator 7812. C2 is a decoupling capacitor commonly used at the output of the voltage regulator.

Construction and testing

An actual-size PCB layout of the electronic thermostat circuit for a fridge is shown Fig. 2 and its component layout in Fig. 3.

Sensor transistor 2N2222 should be mounted on the side wall of the freezer with some epoxy (like Araldite). Connections from the sensor to the circuit should be made with a shielded cable made of materials like polycab coatings. Leads should be fully insulated from each other and from the transistor body using some epoxy so that water

PARTS LIST

Semiconductors:

- IC1 - LM339 quad comparator
- IC2 - 7812, 12V regulator
- D1-D5 - 1N4001 rectifier diode
- ZD1 - 5.2V Zener diode, 400mW
- T1 - 2N2222 transistor as sensor
- T2 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 470-ohm
- R2 - 47-kilo-ohm
- R3 - 15-kilo-ohm
- R4 - 33-kilo-ohm
- R5 - 1.2-kilo-ohm
- R6 - 4-kilo-ohm
- R7 - 10-mega-ohm
- R8, R9 - 10-kilo-ohm
- R10 - 220-ohm
- VR1 - 1-kilo-ohm trimpot

Capacitors:

- C1 - 470 μF , 35V electrolytic
- C2, C4, C5 - 0.1 μF ceramic disk
- C3 - 1 μF , 25V electrolytic

Miscellaneous:

- CON1 - 2-pin plug for AC mains
- CON2-CON4 - 2-pin connector terminal
- X1 - 230V AC primary to 15V, 300mA secondary transformer
- S1 - On/off toggle switch
- F1 - 0.5A fuse
- RL1 - 12V, 1 C/O relay

deposition on the sensor does not short these.

I housed the sensor's metallic body in a ceramic block and affixed it to the freezer's body. This fitting is quite tight to ensure complete thermal contact. Ceramic is the best thermal conductor and electrical insulator.

Calibration is done by bringing the sensor in contact with ice water and at the same time biasing its base emitter junction at about 100 μA using a series resistance and suitable voltage. Junction voltage thus observed corresponds to 0°C . Junction voltage corresponding to any other temperature can then be determined from the figure of temperature coefficient ($-5\text{mV}/^{\circ}\text{C}$).

For troubleshooting, ensure various test points are as per the voltages listed in the test point table. **EFY**



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